What Do Gains in Personalised Reward Modelling Measure?

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Abstract

Personalisation gains can reflect improved rating calibration, predictable differences between users, or better choices than a population model. We evaluate these separately using held-out users and prompts, few-shot adaptation, population baselines and exchanged-parameter controls. Across three preference and safety datasets, text-independent rater offsets improve graded-label log-likelihood by 0.08–0.10 nats per rating from twenty ratings, while our compressed-feature reimplementations of published methods show small direction or choice gains on MultiPref and native PRISM choices. On Community Alignment, twenty support choices from other prompts predict which of two disagreeing users selected which response on identical four-response items: assignment accuracy is 62.2 % [58.6, 65.6], against 50 % chance. A predictor using response positions alone is also informative, and we do not separate content from position. On the shared-item panel, isotropic adaptation gains 0.70 pairwise accuracy points over the population model, although the user-by-item interval includes zero; log-likelihood improves under both bootstrap analyses. The frozen procedure does not establish benefit on MultiPref, and frozen first-turn deviations improve accuracy on users' later, disjoint conversations under user clustering but do not meet the prespecified joint accuracy-and-likelihood criterion over a history-conditioned population model. Changing the population-training procedure shifts measured gains by about 0.6 points. These findings motivate separate evaluations of rating calibration, user-associated choice predictability and predictive benefit over a population baseline.

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1. Introduction

Reward models and LLM judges are trained on human ratings that disagree, and a recent line of work treats this disagreement as signal: distributional preference learning (Siththaranjan et al., 2024), jury learning (Gordon et al., 2022) and personalised reward models that infer a per-user latent from a few examples (Poddar et al., 2024; Chen et al., 2025), within the pluralistic-alignment agenda (Sorensen et al., 2024).

What these models learn about a person is rarely tested directly. Evaluations typically use synthetic users whose preferences differ by construction, binary preference labels, or users seen in training. We ask a simple question under a common protocol on real raters: **given k ratings from someone the model has never seen, how much better can it predict their other ratings — in particular, which response they prefer — and what is it that it learns?** We find that three outcomes usually reported together come apart: the largest few-shot gains on graded ratings are captured by text-independent scale-use offsets; users' choices among identical alternatives are predictable from their other choices; and improvements in choice over a competent population model are small, dataset-dependent and sensitive to how that population model is trained. Semi-synthetic controls characterise which injected content-preference families the evaluated methods recover.

Measurement theory suggests a candidate answer that reward modelling largely ignores: raters differ in **how they use the response scale** — some say "clearly better" where others say "slightly better", some rarely call a tie, some are harsher across the board (Linacre, 1989; Baumgartner & Steenkamp, 2001). Such response styles are text-independent, cheap to estimate, and invisible to models that condition only on content or demographics. We make them a reference point for calibration. On two-sided preference scales, symmetric offsets mainly model intensity and ties and cancel exactly for binary A/B labels at a fixed text head; unrestricted offsets can also change predicted decisions. Likelihood gains alone therefore do not establish content-dependent personalisation, and content adaptation should be evaluated with rater offsets retained.

Contributions. (1) **Graded-rating gains can be scale use.** Under a new-rater protocol (held-out raters and prompts,

k-shot adaptation, population and exchanged-parameter controls), a few text-independent rater offsets capture the largest demonstrated few-shot gains on three preference and safety datasets and two code-readability studies; in their symmetric form they chiefly improve graded-label likelihood, change direction on graded MultiPref only slightly, and cancel exactly under binary labels (§4–5). (2) **User differences need their own test — ideally on identical items.** On Community Alignment, where many users judged the same prompt with the same four responses, support histories predict which user chooses which response among identical alternatives (62 % assignment accuracy between disagreeing users against 50 %); attribute-derived scores remain predictive when attributes and response positions are fitted jointly, and a position-only predictor is also informative (§6). (3) **Beneficial personalisation needs a competent population baseline.** Own-versus-other contrasts do not measure benefit; gains over the population model are small (+0.3 to +0.7 choice points where established), not established on MultiPref, and the measured gain changes by ˜0.6 points with the population-training procedure alone (§6). Semi-synthetic positive controls characterise the evaluated learners' recovery of specified injected preference families (§5.6); a screened attribute learner and the covariance-calibrated procedure are reported as supporting tools (Appendix G, F). Protocols, amendments and failed hypotheses are in Appendices A, B, E and F.

2. Related work

Pluralistic and distributional reward models. Distributional preference learning (Siththaranjan et al., 2024), diverging-preference detection (Zhang et al., 2025), PRISM (Kirk et al., 2024), MultiPref (Miranda et al., 2025) and DICES (Aroyo et al., 2023) motivate modelling disagreement; jury learning (Gordon et al., 2022), multi-annotator models (Mostafazadeh Davani et al., 2022), disagreement deconvolution (Gordon et al., 2021), per-rater toxicity models (Kumar et al., 2021), opinion-based prediction of unseen annotators (Fleisig et al., 2023) and multilevel models of DICES raters (Homan et al., 2024) predict individual annotators. Preference strength enters reward training as a margin (Touvron et al., 2023; Wang et al., 2025); ties have classical paired-comparison models (Rao & Kupper, 1967; Davidson, 1970). **Personalised reward models** infer a user latent from a few examples (Poddar et al., 2024; Chen et al., 2025), use a low-rank reward subspace (Bose et al., 2025), persona prompts induced from history (Ryan et al., 2025), per-user embeddings (Li et al., 2024), combinations of base rewards (Shenfeld et al., 2025; Luo et al., 2025) or neural processes (Kobalczyk et al., 2024); others personalise the policy (Jang et al., 2023; Feng et al., 2024), adapt to groups (Zhao et al., 2024; Chakraborty et al., 2024), study personalisation versus aggregation (Park et al., 2024) or condition a decoder on value profiles of held-out raters (Sorensen et al., 2025). Benchmarks often simulate users (Castricato et al., 2025; Zollo et al., 2025); real multi-rater data with diverse candidates are emerging (Zhang et al., 2026). VPL's latent can collapse under sparse data (Kim & Kim, 2026), and methods differ widely in a multi-faceted comparison (Dong et al., 2025). Rater groups use safety severity scales differently (Mishra et al., 2026); demographics are often weak predictors (Orlikowski et al., 2023). **Measurement theory and annotator models.** Rater severity and scale use are standard facets of many-facet Rasch measurement (Linacre, 1989; Eckes, 2011) and response styles a classical survey topic (Baumgartner & Steenkamp, 2001); user and item biases carry much of the gain in collaborative filtering (Koren et al., 2009), user-specific ordinal thresholds predict personal rating distributions (Koren & Sill, 2011), and annotator models estimate rater-specific labelling behaviour (Dawid & Skene, 1979; Raykar et al., 2010; Hovy et al., 2013; Deng et al., 2023; Plank, 2022; Uma et al., 2021). Our offsets are of this kind and are not new; what is new is testing personalised reward models against them on new raters and asking what these models could detect.

3. Data, and where the predictable structure sits

Data. MultiPref (Miranda et al., 2025; 10,461 comparisons; 4 ratings each by 2 crowd and 2 expert raters; 227 persistent raters; 5 ordered levels A-clearly … B-clearly). PRISM (Kirk et al., 2024; 68,371 responses each rated 1–100 by one of 1,396 users; binned to 5 fixed levels). DICES-350 (Aroyo et al., 2023; 350 conversations fully crossed with 104 quality-checked raters; safe / unsure / unsafe).

A learner-free measurement (Appendix D). Before fitting any personalised model we measured, on MultiPref, where held-out predictable structure sits: beyond item location, rater-specific thresholds improve held-out likelihood about 25 times more than item-specific dispersion (+0.036 vs +0.0014 nats per rating), the item-level shape of the rating distribution that replicates across disjoint rater panels is small (split-half reliability 0.25), and about a tenth of it is decodable from text. Beyond item location, the predictable structure is therefore mostly rater-side scale use, while the item-level shape that replicates across disjoint panels is small and only partly in the text. These two facts come from different estimands and do not say that disagreement is "not in the text". The stage-1 decision rule we had written down returned **"mixed"**, because G/X was not ≤ 10 % (interval 5.7–11.9 %); the factorised model of §4 was chosen with that outcome known.

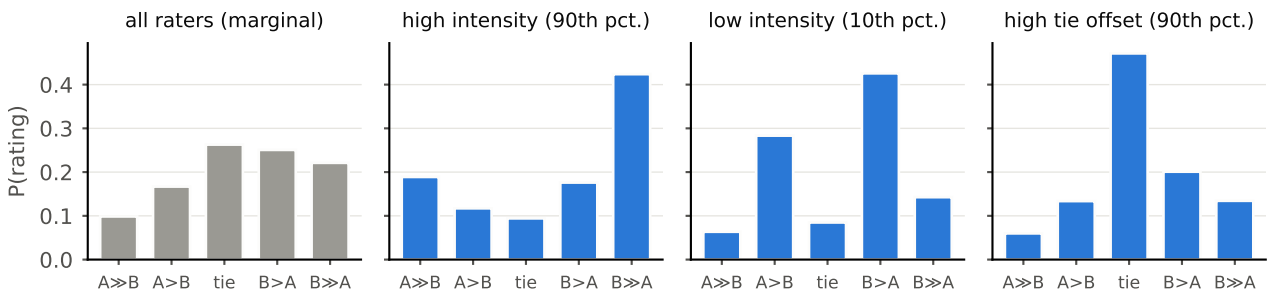


Figure 1. What three numbers per rater change. Left: the marginal distribution of MultiPref ratings. Right: the same distribution passed through the fitted scale-use offsets (extreme, slight, tie) of three real new raters (k = 50) closest to the 90th / 10th percentile of intensity (extreme minus slight) among raters whose tie offset is at or below the median, and closest to the 90th percentile of the tie offset. The offsets are text-independent: they reshape how strongly a rater is predicted to prefer, not which side.

4. A three-parameter scale-use baseline

(Figure 2 illustrates the effect of the offsets.) Let $\ell(x) \in R\hat{}K$ be the class logits of a text-only distributional reward model for item x (K ordered rating levels). For rater r we add a text-independent offset vector: $p(y \mid x, r) = \text{softmax}(\ell(x) + b_r)$. Variants: **DF** ($b_r \in R\hat{}K$, free), **DFs** (two-sided preference scales only: b_r tied symmetrically, (e, s, t, s, e) — extreme, slight, tie — 3 parameters, of which 2 are identifiable because the softmax is invariant to a common shift; no A-vs-B position term). Rater offsets have a Gaussian prior centred on the training-rater mean; a new rater's offsets are the MAP estimate from k of their ratings with the text model frozen. Baselines under the same protocol: **D** (text-only distributional RM), **DG** (group offsets: crowd/expert, race, region), **DL** (low-rank text-interacting latent: $\ell(x) + (Wx + w_0) \cdot z_r$, d = 8, the personalised-RM family). An interpretable probit family (rater shift / log-scale on an ordered-probit text head) is reported as a secondary analysis.

Training and adaptation. The text model and the training raters' offsets are fitted jointly by penalised maximum likelihood,

$$\min \sum_{(i,r,y)} -\log p(y \mid x_i, r) + \lambda_{\text{text}} \|V\|^2 + \lambda_r \sum_r \|b_r\|^2,$$

with λ_{text} chosen on the text-only model and λ_r on held-out items of the training raters. A new rater r^* is adapted from k ratings $\{(x_j, y_j)\}$ by $\hat{b}_{r^*} = \arg\min_b \sum_j -\log \text{softmax}(\ell(x_j) + b)_{y_j} + \lambda_r \|b - \bar{b}\|^2$, with $\bar{b}$ the training-rater mean (the prior mean; $\hat{b} = \bar{b}$ at $k = 0$) and the text model frozen — a convex problem in 3–5 parameters, solved in milliseconds. Any text model that outputs graded log-probabilities can be used: we use a ridge head on frozen LLM features (§5.1–5.2), a LoRA-fine-tuned LLM reward model (§5.3) and the decoder of a published personalised reward model (§5.4).

5. Experiments

Protocol. 5 folds; in each, 20 % of raters (stratified by group) and 20 % of prompt groups are held out; models are trained on training raters × training items with penalties chosen on a validation split; each test rater is adapted on $k \in \{0, 5, 10, 20, 50\}$ of their ratings of *training* items (5 random draws) and evaluated on their ratings of *held-out* items. Primary metric: held-out log-likelihood per rating; also RPS and preference-direction accuracy; rater-cluster bootstrap (B = 2,000). Because test raters share test items on MultiPref and DICES, the headline k = 20 contrasts were also re-estimated with a two-way bootstrap that resamples raters and prompt groups independently: on graded labels every log-likelihood interval widens by at most 0.007 on either side and no log-likelihood conclusion changes; direction-accuracy differences of about a point or less (the style model's +0.5 on MultiPref, VPL's direction advantage over the style model) lose significance and are marked where they appear. Text model in §5.1–5.2: frozen Qwen2.5-7B features with a ridge head; §5.3: a LoRA-fine-tuned reward model.

5.1. Main result (MultiPref)

Figure 1 shows the gain over the text-only reward model as a function of k on all three datasets. In the table, DF/DFs/DG use the training-rater prior and DL and the DFs − DL column use the few-shot-tuned prior (the DFs − DL contrast is post hoc).

Preference-direction accuracy (D: 0.674): DFs +0.005 [+0.001, +0.008] at k = 20 (not significant once items are also resampled: [−0.001, +0.011]); DF −0.014 [−0.020, −0.007]. A three-number style model gains as much as a tuned 8-dimensional text-interacting latent (tie at k = 20) and more at k = 5, 10 and 50; fitting a rater's A-vs-B position from few ratings costs direction accuracy.

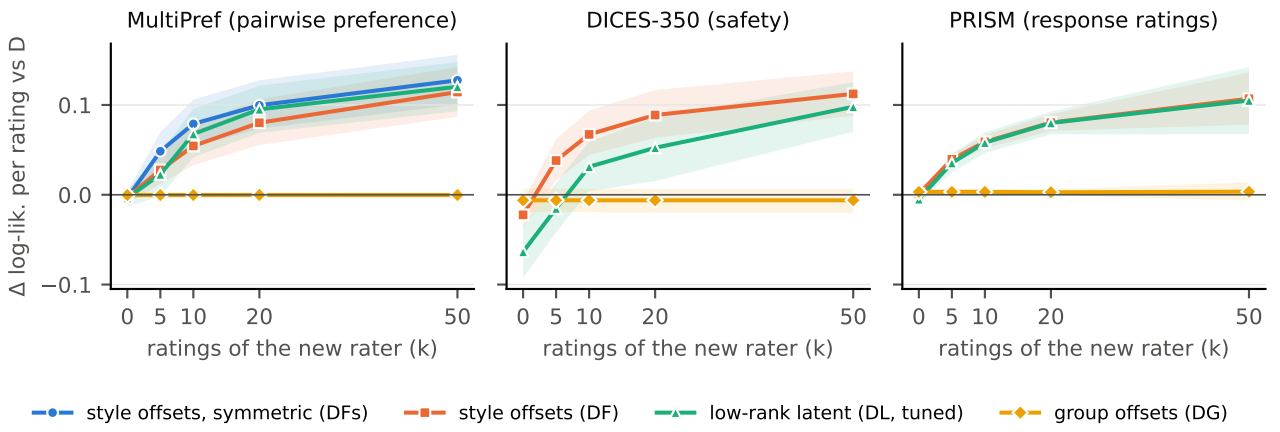


Figure 2. New raters: gain in held-out log-likelihood per rating over the text-only distributional reward model (D), as a function of the number k of the new rater's ratings used for adaptation; 95% rater-cluster bootstrap bands, 5 folds. Text-independent rater offsets (DF; symmetric three-parameter DFs on two-sided preferences) gain 0.08–0.10 nats per rating at k = 20 on all three datasets; group offsets (crowd/expert, race, region) gain nothing; a low-rank text-interacting rater latent with a few-shot-tuned prior ties the three-parameter DFs and beats the five-offset DF on MultiPref (DF − DL = −0.015). DF, DFs and DG use the training-rater prior; DL the few-shot-tuned prior.

k	DF − D	DFs − D	DG − D	DL − D (tuned prior)	DFs − DL (tuned, post hoc)
5	+0.028 [+0.012, +0.044]	+0.048 [+0.029, +0.070]	≈ 0	+0.023 [−0.002, +0.047]	+0.026 [+0.016, +0.037]
20	+0.080 [+0.055, +0.106]	**+0.100 [+0.074, +0.128]**	≈ 0	+0.095 [+0.069, +0.122]	+0.0045 [−0.0004, +0.0092]
50	+0.114 [+0.087, +0.142]	+0.128 [+0.101, +0.156]	≈ 0	+0.120	+0.007 [+0.002, +0.013]

Reference points. To size these gains: on MultiPref at k = 20 a uniform forecast scores −1.609 nats per rating, a forecast that ignores the text and uses only the new rater's 20 ratings −1.487, the text-only model −1.441, the text model with the three offsets −1.342, and offsets fitted in-sample on all of the rater's held-out ratings (an optimistic ceiling for text-independent offsets) −1.262. Twenty ratings of a person, without the text, recover 73 % of the text model's gain over uniform; the offsets close 56 % of the gap between the text-only model and the in-sample ceiling. On DICES the rater's ratings alone predict better than the text alone (−0.766 vs −0.840 at k = 20). Even the in-sample ceiling raises MultiPref direction accuracy by only 1.3 points (0.674 → 0.687), confirming that text-independent offsets cannot carry ranking.

5.2. Replications (k = 20)

DF − D and DG − D use the training-rater prior, DF − DL the few-shot-tuned prior. Direction accuracy on PRISM means good-vs-bad around the scale midpoint.

5.3. On a fine-tuned reward model

With a LoRA-fine-tuned Qwen2.5-7B reward model as the text model, the three offsets still add +0.100 [+0.071, +0.130] nats per rating at k = 20 — as on the frozen-feature head — and the gains of fine-tuning and of personalisation add. Against in-context personalisation (a LoRA model given the rater's k ratings in the prompt, three seeds per fold) the offsets are better by +0.041 [+0.027, +0.054] nats and +1.9 [+0.9, +2.7] direction points (Appendix B).

5.4. Published personalised reward models

We reimplemented VPL (Poddar et al., 2024) and PAL-B (Chen et al., 2025) from their official code as heads on the same frozen per-response embeddings and evaluated them under the same new-rater protocol, refitting the text-only and style models on those embeddings. On graded labels, at k = 20 the style model is better than VPL by +0.092 [+0.068, +0.119] nats per rating ([+0.063, +0.124] when prompt groups are also resampled) and than PAL-B by +0.158. VPL's latent does carry rater information once its posterior collapse is prevented (KL weight selected as 0 or 0.001; +0.087 over its own unpersonalised version), but only lifts it to the level of the text-only model; PAL-B's few-shot user weights give no gain. Their user representations are unrelated to response style (cross-validated $R^2 \approx 0$ against the style offsets). On binary labels — the setting in which both methods were proposed — every method's personalisation gain is at most 0.002 nats (small but nonzero for PAL-B with K = 4, and for VPL and the style model at k = 50), and

dataset	DF − D	DG − D (groups)	DF − DL (tuned prior)
DICES-350 (safety)	+0.089 [+0.064, +0.117]	−0.006 (race)	+0.037 [+0.022, +0.053]
PRISM (ratings)	+0.080 [+0.071, +0.090]	+0.003 (region)	−0.001 [−0.006, +0.003] (formally not non-inferior: bound $-0.006 < -0.005$)

VPL's text head is the best binary model (the style model is 0.010 nats behind it and 0.005 behind the text-only model, differences of the size of item-sampling noise): on these panels the largest demonstrated gains are captured by calibration of graded levels and ties, which binary preference data discard. VPL's per-response reward head is slightly better at predicting *which* response is preferred (0.677 vs 0.668), an effect present without personalisation and not significant once prompt groups are also resampled (style model − VPL: −0.9 points [−2.7, +0.8]). The offsets are also a plug-in: added to VPL (fitted on the same k ratings, VPL frozen) they improve it by +0.088 [+0.066, +0.111] nats per rating at k = 20, tie with the style model on likelihood (−0.004 [−0.014, +0.005]) and keep VPL's direction accuracy (+1.1 points over the style model alone; rater bootstrap only).

5.5. Summary of baselines (MultiPref, new raters, k = 20)

Gains are relative to each block's text-only model (D, Dp, D-LoRA); the LoRA block uses adaptation draw 0 only (the draw the in-context model was evaluated on), hence the style model's +0.092 here against +0.100 averaged over five draws in §5.3.

Rows within a block share the text model and the raters; across blocks the text models differ (text-only direction accuracy 0.674 / 0.670 / 0.688). VPL's and PAL-B's own gains over their unpersonalised versions are +0.087 and −0.000.

5.6. Could these methods have detected content personalisation? A semi-synthetic positive control

A ranking null is informative only if the methods would detect content-dependent preferences where they exist. We keep MultiPref's raters, items, folds and k-shot draws and replace the labels by semi-synthetic ones: a generator (the text-only head plus scale-use offsets) gives each rating a distribution, to which a rater-specific content term $(\theta_r^\top g_i)\,(-2,-1,0,1,2)$ is added; g_i are two response-pair features (log length ratio and a random embedding direction orthogonal to consensus quality), $\theta_r \sim N(0, \tau^2 I)$, and τ is calibrated so that an oracle knowing θ_r gains +1.7, +5.2, +10.1 or +21.1 direction points ($\tau = 0$: null control). The contrast is **own adapted parameters minus another test rater's** (a derangement within fold, k and draw). We fixed beforehand that the protocol has power at a strength if some content-interacting method has a lower 95 % bound above +1 point at k = 20; the +10 and +21 panels are exploratory.

Own minus another rater's k = 20 ratings, direction accuracy in points, 95 % rater bootstrap; 209 new raters, 5 folds. The low-rank latent uses the few-shot-tuned prior. k = 50 contrasts use the 180 of 209 raters with at least 50 training-item ratings. PAL-B's own − other contrast is negative with compressed features (null control −0.23, real data −0.43): its per-rater adaptation adds noise there; with full features it is at zero (−0.04, +0.11). The injection uses one draw of θ and of the labels, so seed variability of the synthetic panels is not in the intervals. Last column: PAL-B on the full 3,584-dimensional features instead of their PCA-256 (exploratory; $\tau 2$ not run).

No method has power at the pre-specified strengths. In the exploratory dose-response, VPL and PAL-B detect +21-point preferences but recover less than half a point of +10; on real data they are at the null-control level, and the low-rank latent's +0.30 [−0.00, +0.60] is matched by the text-independent offsets (+0.32). A reference learner that is given the two injected features recovers +0.89 [+0.40, +1.37], +3.1 and +8.0 points of the +1.7, +5.2 and +10.1-point oracle gains from 20 ratings, so twenty ratings permit recovery in these injected families when the generating features are supplied; the gap implicates representation and inductive bias (probes, re-injection and semantic axes in Appendix B). These results characterise recovery for specified injected families on panels noisier than the real data; they are not formal exclusion bounds.

5.7. How much room is there? Conflicts among aspect judgements are rare

With the rater's own helpfulness, truthfulness and harmlessness labels as inputs, a pooled trade-off gets 97.5 % of MultiPref directions right and a rater-specific one adds only +0.10 [−0.24, +0.43] points: the three aspect judgements conflict in only 4.0 % of held-out comparisons (details in Appendix B).

5.8. A learner that finds the axes

A screened empirical-Bayes learner over 24 interpretable attribute differences (SAP) recovers the semi-synthetic content preferences that the published learners miss, with near-zero effects on null panels, and abstains on real MultiPref under a prospectively specified rule (+0.28 [−0.00, +0.58] points at k = 20). Details: Appendix G.

text model	personalisation	gain in LL per rating vs the block's text-only
frozen features (joint item embedding)	group offsets (crowd / expert)	≈ 0
	low-rank latent, tuned prior	+0.095 [+0.069, +0.122]
	style offsets, 3 numbers (DFs)	**+0.100 [+0.074, +0.128]**
frozen features (per-response embeddings)	VPL (reimplemented)	+0.002 [−0.008, +0.012]
	PAL-B (reimplemented)	−0.065 (worse)
	DFs	**+0.094 [+0.069, +0.120]**
	VPL + style offsets	+0.090
LoRA-fine-tuned Qwen2.5-7B (209 raters)	in-context rater history (RC-LoRA, mean of 3 training seeds)	+0.051 [+0.024, +0.079]
	DFs	**+0.092 [+0.063, +0.125]**

panel	oracle gain	low-rank latent (DL)	VPL	PAL-B	PAL-B, full features
null control ($\tau = 0$)	0	−0.10 [−0.38, +0.17]	−0.04 [−0.08, +0.01]	−0.23 [−0.45, +0.00]	−0.04 [−0.25, +0.17]
injected, $\tau 2$	+1.7	−0.01 [−0.29, +0.26]	+0.01 [−0.07, +0.10]	−0.10 [−0.36, +0.15]	—
injected, $\tau 5$	+5.2	−0.02 [−0.28, +0.26]	+0.01 [+0.00, +0.03]	+0.23 [−0.04, +0.51]	+0.26 [−0.14, +0.66]
injected, $\tau 10$ (exploratory)	+10.1	+0.09 [−0.18, +0.36]	+0.27 [+0.02, +0.54]	+0.42 [−0.08, +0.96]	+1.95 [+1.16, +2.74]
injected, $\tau 20$ (exploratory)	+21.1	+1.47 [+0.83, +2.18]	+6.92 [+5.63, +8.30]	+4.62 [+3.47, +5.78]	+8.13 [+6.54, +9.77]
real MultiPref	—	+0.30 [−0.00, +0.60]	−0.01 [−0.04, +0.02]	−0.43 [−0.72, −0.15]	+0.11 [−0.16, +0.37]

5.9. Developers rating code readability

On two code-readability studies with persistent developer identities (Buse–Weimer, 121 raters; Dorn, 3,393 developers) and Qwen2.5-Coder-7B features, scale use transfers to new developers (offsets +0.054 [+0.036, +0.074] nats per rating at k = 20 and +0.040 [+0.031, +0.049] at k = 10), while content adaptation over 22 readability attributes is inconclusive at the pre-set k = 20 (+0.40 [−0.25, +1.06] and −0.19 [−0.81, +0.42] concordance points; Appendices A–B).

6. What personalisation buys: graded-rating prediction, user signal, and small choice gains

Binarisation removes the symmetric scale-use term; evaluated adaptation gains on MultiPref are small. On MultiPref with ties dropped and labels binarised, every evaluated method's personalisation gain is at most 0.002 nats per rating (§5.4), and the scale-use baseline's binary accuracy is essentially unchanged from the text-only model's. On these graded panels the largest demonstrated gains concern graded levels and ties — which binary preference data, the format in which most personalised reward models are trained and evaluated, discard (Community Alignment shows that binary choices can also carry predictable user differences, §6). **Ranking moves little.** Symmetric offsets mainly change how strongly a rater is predicted to prefer a response: on graded MultiPref the scale-use baseline raises direction accuracy by only +0.005 [+0.001, +0.008] at k = 20 (about half of it already at k = 0), while RPS drops from 0.1593 to 0.1532. On single-response safety ratings the same offsets do move the safe/unsafe decision (+4.8 points on DICES), because there a rater's threshold *is* the decision. Text-only direction accuracy on MultiPref (0.674) is already as high as the agreement in direction between two human raters of the same item (0.673 [0.666, 0.680]): a population model predicts a new rater about as well as another person does. **Seen raters.** With all their training ratings (median 134), training raters' own parameters beat another rater's by −0.11 [−0.57, +0.36] direction points for the low-rank latent (offsets +0.10, −0.23): the ranking null is not specific to few-shot adaptation (exploratory, Appendix B). Against its own k = 0 version, no method's direction accuracy on MultiPref rises by more than 0.2 points at k = 20, and every upper 95 % bound stays below +0.8 points even when prompt groups are resampled together with raters (VPL +0.07 [−0.57, +0.74], style model −0.10 [−0.69, +0.54], PAL-B −0.48 [−1.11, +0.14]). **The ranking null replicates on real users' own choices.** PRISM users each saw 2–4 responses to their own prompt, scored them and chose one; pairing the chosen response with every other one gives 41,199 native pairwise choices of 1,396 users. Under the same new-user protocol (5 user folds × 5 conversation folds, k of the user's pairs from training conversations), we fixed before any result that the ranking null would be stated only if every method's improvement in choice accuracy over its own k = 0 version had an upper 95 % bound below one point at k = 20. It holds for all of them (1,033 new users at k = 0; the 669 with at least 20 adaptation pairs at k = 20): VPL +0.12 points [−0.08, +0.33], PAL-B +0.0 [−0.4, +0.4] (K = 2) and +0.3 [−0.1, +0.7] (K = 4), the 5-offset model −0.6 [−1.4, +0.2]; the symmetric scale-use model is identical to the text-only model on binary choices, because a binary label carries no intensity. LoRe (Bose et al., 2025), which reports +10 points for unseen PRISM users, run here with its own few-shot fit and with a validated adaptation ridge under the same pre-set rule, also fails it: own gain at k = 20 −1.5 [−2.5, −0.4] (regularised −0.5 [−1.2, +0.2]), own − other

+0.2 [−0.8, +1.2]. In LoRe's own setting (its embeddings, split and pair construction) we obtain 67.4 % for the monolithic model and 63.6 % for LoRe (published 61.0 % and 71.0 %; Appendix B): the published gain is not reproduced. With a graded label cut from the users' own score differences (thresholds fixed in advance), the MultiPref pattern returns at half the size (scale-use model +0.043 [+0.031, +0.057] nats per rating at k = 20, VPL +0.015, PAL-B 0.000; no direction gain above +0.2 points). **Community Alignment: who chooses what on identical items, and what adaptation buys.** Community Alignment (CA; first turns; 2,957 new users in five countries and languages; each choice event's chosen response paired with each of the other three). Accuracy below is pairwise winner-versus-other accuracy, not four-way winner selection. Status: the adaptation procedure was developed exploratorily on first turns (Appendix F); the Stage-12 and Stage-13 outcome rules were frozen before their outcomes; the Stage-13 first-turn analyses reuse previously examined data, while the second- and third-turn outcomes were evaluated for the first time.

Identity on identical items (Stage 13, frozen). CA contains 2,928 exact prompt–four-response items shown to at least two users, covering 49,616 first-turn conversations (median 15 users per item, one language per item, fixed response order). For two users of the same item who chose different responses a and b, the population utilities cancel and the adapted models predict the assignment from $S = (\delta_u - \delta_v)^\top (x_a - x_b)$, with model-implied probability $\sigma(S)$. With validated isotropic adaptation over length, paragraphs and list/heading lines (k = 20 support events on other prompts; the Stage-12 procedure unchanged), the **conditional assignment log score gains +0.066 [+0.034, +0.091] nats over chance and assignment accuracy is 62.2 % [58.6, 65.6]** (4,608 disagreeing pairs from 1,336 users on 1,116 items; item × user bootstrap). It is unchanged with uncentred features; positive-multiplier and prompt-clustered intervals are [+0.046, +0.083] and [+0.036, +0.093]; every fold is positive. The estimand is narrow: given two disagreeing users, their support choices and the two responses they picked, assign the responses; it conditions on the fitted deviations and population model.

Position. The four fixed positions differ systematically in these attributes (label-blind check), so we refitted with position indicators added (after the result). The attribute-derived part of the jointly fitted scores keeps +0.060 [+0.032, +0.084] nats, a position-only predictor is also informative (63.8 %), and the joint model's paired gain over position-only is +0.026 [−0.0002, +0.046], so additional assignment value from attributes beyond position is unresolved. With correlated predictors under one prior this does not separate content from position: we claim user-specific choice predictability on identical alternatives, not a content preference, and not a psychological position habit.

Benefit over the population model. On all test events of the shared items in the test folds — a broader panel than the 1,116 items that contribute disagreeing pairs to the assignment test (2,383 items, 1,436 users; population accuracy 72.4 %) — isotropic adaptation gains **+0.70 [+0.24, +1.14] pairwise accuracy points and +0.018 [+0.014, +0.023] nats** — the pre-set user-bootstrap rule for benefit is met, but the user × item interval for accuracy covers 0 ([−0.07, +1.43]) while log-likelihood improves under both, so an accuracy benefit that generalises across items is unresolved; nor is the gain shown to be below one point. Evaluated frozen on the same users' unanalysed second turns (procedure refitted there; Stage 12), the covariance-calibrated variant gains +0.38 [+0.13, +0.64] points and +0.0046 nats and isotropic +0.31 [+0.05, +0.58]; the prompt × user interval covers 0. On MultiPref the frozen isotropic procedure did not establish improvement (−0.28 [−0.64, +0.08] points, 202 held-out annotators). On CA the estimated covariance shows no advantage over isotropic shrinkage (turn 1 −0.17 [−0.29, −0.05], turn 2 +0.07 [−0.07, +0.22] points); on MultiPref it beats isotropic under the rater bootstrap (+0.23 [+0.04, +0.42]) without establishing improvement over the population model (two-way includes 0; Appendix F). Validated scales imply strong shrinkage, and realised gains are far below model-implied ones (+1.9 points, Gaussian event-level simulation).

Transport to later conversations (Stage 13, frozen). We froze each user's first-turn deviation and added it, without rescaling, to a third-turn population model that sees the conversation history (the earlier prompts and the user's preferred responses). We scored it on the third turns of conversations not used for support. Of 1,901 users with a frozen first-turn deviation, 1,808 have at least five third-turn events with a complete history and 953 have at least five outside their support conversations. The result is accuracy-positive under user clustering and likelihood-inconclusive: accuracy rises by +0.43 [+0.13, +0.74] points (user × prompt interval [−0.08, +0.92]), log-likelihood by +0.0015 [−0.0006, +0.0035] nats. The pre-set rule requires both lower bounds above 0, so additional predictive value of unrescaled first-turn coefficients beyond a history-conditioned population model is **not supported for this procedure and panel**. This neither shows that transport has no effect nor that preferences are unstable, and it does not test other ways of carrying information across turns.

The measured gain depends on the population baseline. On 1,799 common users, one-dimensional length adaptation gains +0.88 / +1.08 points (random / prompt-separated test events) against a population model fitted on low-volume annotators, as in WIMHF's protocol, and +0.26 / +0.47 against one fitted on all other-fold users: the population-procedure effect is +0.62 [+0.41, +0.81] and +0.61 [+0.30, +0.93] (paired), while the prompt-split contrast is inconclu-

Table 1. Choice-level results. k = 20 support choice events per user (20 binary comparisons on MultiPref); pairwise accuracy; log-likelihood per pair. First interval: user (rater) bootstrap, or item × user for the assignment rows; second: prompt- or item-group × user two-way bootstrap. Nominal 95 % intervals, conditional on the fitted population models. "Reused" = data examined in earlier stages; "new" = outcomes evaluated for the first time.

Question (comparator)	Panel	Accuracy, points	Log-likelihood, nats	Status
Assignment (chance)	CA t1 identical items; 1,336 users	62.2 % [58.6, 65.6]	+0.066 [+0.034, +0.091]	frozen, reused
… beyond position (position-only)	same	—	+0.026 [−0.0002, +0.046]	exploratory
Benefit (population)	CA t1 shared items; 1,436 users	+0.70 [+0.24, +1.14]; [−0.07, +1.43]	+0.018 [+0.014, +0.023]; [+0.010, +0.026]	frozen, reused
Benefit, isotropic (population)	CA t2; 1,702 users	+0.31 [+0.05, +0.58]; [−0.13, +0.79]	+0.0038 [+0.0021, +0.0057]; [+0.0006, +0.0066]	frozen, new
Benefit, TRV (population)	CA t2; 1,702 users	+0.38 [+0.13, +0.64]; [−0.04, +0.83]	+0.0046 [+0.0031, +0.0063]; [+0.0018, +0.0074]	frozen, new
Benefit, isotropic (population)	MultiPref; 202 annotators	−0.28 [−0.64, +0.08]; [−0.92, +0.39]	−0.0004 [−0.0019, +0.0010]; [−0.0028, +0.0019]	frozen, reused
Transport (history-conditioned pop.)	CA t3; 953 users	+0.43 [+0.13, +0.74]; [−0.08, +0.92]	+0.0015 [−0.0006, +0.0035]; [−0.0020, +0.0048]	frozen, new
Baseline procedure W − O (paired)	CA t1; 1,799 users	+0.62 [+0.41, +0.81]	—	exploratory

sive (+0.20 [−0.20, +0.59]). The two procedures differ in volume, composition and prior estimation, so no mechanism is isolated. (Our line-count proxy is not WIMHF's discovered feature; with it, list/heading adaptation does not beat the population coefficient — a related protocol study, not a failed reproduction.)

Earlier evidence. Before this test, own adapted parameters beat another user's for four of six methods (LoRe +3.27 [+2.85, +3.70]; within-country +3.22) and length deviations reproduced across disjoint prompt sets; own − other does not measure benefit and task mix could create it (Appendix F). The identical-item test supersedes these as evidence of user-specific choice predictability.

7. Recommendations

(i) **Separate three outcomes.** Report graded-rating likelihood (calibration and resolution), a test of user-specific choice predictability, and choice or ranking gains over the population model as different results; a gain on one does not imply the others. (ii) **Test user differences on identical items** where the data allow it (an assignment test between disagreeing users, as in §6): own-versus-other contrasts on different prompts confound user differences with task mix and do not measure benefit. (iii) **Report a competent population baseline and its training procedure.** The measured personalisation gain changed by about 0.6 points with the population-training procedure (§6). (iv) **Report a scale-use baseline** (a few text-independent offsets per rater) next to any personalised reward model on graded labels, and evaluate content adaptation with those offsets retained. (v) **Evaluate on raters unseen in training and on held-out prompts**, with permutation and positive controls. (vi) Keep graded labels and ties when individual differences matter.

8. Discussion

What do personalisation gains measure? Three things often reported as one come apart. Graded-rating gains for new raters are largely scale use: offsets capture them, probes find near-zero R^2 for them in the evaluated user representations, and adding offsets to VPL recovers the whole gain (+0.088). User-specific choice predictability is real on identical CA items (62 %), though not separated from position. Choice gains over a competent population model are small and fragile (Table 1).

Why did published models learn little about new raters here? Published evaluations differ in users (synthetic, or seen in training), label formats, representations and population baselines; the last alone changes the measured gain by about 0.6 points on CA. Unresolved native-reproduction discrepancies prevent attributing the rest to evaluation design, and all method conclusions concern the evaluated implementations.

9. Limitations and analysis log

The main tables use frozen-feature text heads (a LoRA reward model only in §5.3), and compressed features make the published methods weaker detectors than full features. VPL, PAL and LoRe are our reimplementations as heads on frozen embeddings; their published numbers were not reproduced in native settings (LoRe's PRISM gain not even with its own embeddings), for reasons we could not isolate. The positive control injects low-dimensional preferences on MultiPref only, on panels noisier than the real data. The aspect analyses rest on MultiPref's concurrent aspect labels; SAP's limitations are in Appendix G. The first three datasets are English; Community Alignment spans five languages, which the phrase-based attributes cover only for English; and the per-rater calibration is not shown to change a downstream decision. The CA analyses reuse data examined in earlier stages (Appendix A); the identical-item test conditions on fitted models; features omit conversation history except in the third-turn test; and the identity signal is not separated from response position.

Impact Statement

This paper evaluates personalised reward models, which are meant to let language models serve people whose preferences differ. It finds that reported personalisation gains can reflect rating-scale use, user-associated choice predictability or real improvements over a population model, and that these need separate tests: gains on graded ratings were largely scale use, users' choices on identical alternatives were predictable from their other choices, and improvements over a population model were small and dataset-dependent. Predicting individuals' choices also raises privacy and manipulation concerns that a deployment would need to address; we report no individual-level results. The datasets are public, were collected with consent by their creators, and are used under their licences; rater identifiers are used only as anonymous indices.

References

Aroyo, L., Taylor, A. S., Díaz, M., Homan, C. M., Parrish, A., Serapio-García, G., Prabhakaran, V., and Wang, D. DICES dataset: Diversity in conversational AI evaluation for safety. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. doi: 10.52202/075280-2321.

Baumgartner, H. and Steenkamp, J.-B. E. M. Response styles in marketing research: A cross-national investigation. *Journal of Marketing Research*, 38(2):143–156, 2001. doi: 10.1509/jmkr.38.2.143.18840.

Bose, A., Xiong, Z., Chi, Y., Du, S. S., Xiao, L., and Fazel, M. LoRe: Personalizing LLMs via low-rank reward modeling, 2025. URL https://arxiv.org/abs/2504.14439.

Castricato, L., Lile, N., Rafailov, R., Fränken, J.-P., and Finn, C. PERSONA: A reproducible testbed for pluralistic alignment. In *Proceedings of the 31st International Conference on Computational Linguistics (COLING)*, pp. 11348–11368, Abu Dhabi, UAE, 2025. Association for Computational Linguistics. URL https://aclanthology.org/2025.coling-main.752/.

Chakraborty, S., Qiu, J., Yuan, H., Koppel, A., Manocha, D., Huang, F., Bedi, A. S., and Wang, M. MaxMin-RLHF: Alignment with diverse human preferences. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 6116–6135, 2024.

Chen, D., Chen, Y., Rege, A., Wang, Z., and Vinayak, R. K. PAL: Sample-efficient personalized reward modeling for pluralistic alignment. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

Davidson, R. R. On extending the Bradley-Terry model to accommodate ties in paired comparison experiments. *Journal of the American Statistical Association*, 65(329): 317–328, 1970. doi: 10.1080/01621459.1970.10481082.

Dawid, A. P. and Skene, A. M. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 28(1):20–28, 1979.

Deng, N., Zhang, X. F., Liu, S., Wu, W., Wang, L., and Mihalcea, R. You are what you annotate: Towards better models through annotator representations. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 12475–12498, 2023.

Dong, Y. R., Hu, T., Liu, Y., Üstün, A., and Collier, N. When personalization meets reality: A multi-faceted analysis of personalized preference learning, 2025. URL https://arxiv.org/abs/2502.19158.

Eckes, T. *Introduction to Many-Facet Rasch Measurement: Analyzing and Evaluating Rater-Mediated Assessments*, volume 22 of *Language Testing and Evaluation*. Peter Lang, Frankfurt am Main, 2011. ISBN 978-3-631-61350-4.

Feng, S., Sorensen, T., Liu, Y., Fisher, J., Park, C. Y., Choi, Y., and Tsvetkov, Y. Modular pluralism: Pluralistic alignment via multi-LLM collaboration. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 4151–4171, Miami, Florida, USA, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.240.

Fleisig, E., Abebe, R., and Klein, D. When the majority is wrong: Modeling annotator disagreement for subjective tasks. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 6715–6726, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.415.

Gordon, M. L., Zhou, K., Patel, K., Hashimoto, T., and Bernstein, M. S. The disagreement deconvolution: Bringing machine learning performance metrics in line with reality. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*, pp. 1–14. ACM, 2021. doi: 10.1145/3411764.3445423.

Gordon, M. L., Lam, M. S., Park, J. S., Patel, K., Hancock, J. T., Hashimoto, T., and Bernstein, M. S. Jury learning: Integrating dissenting voices into machine learning models. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, 2022. doi: 10.1145/3491102.3502004.

Homan, C. M., Serapio-Garcia, G., Aroyo, L., Diaz, M., Parrish, A., Prabhakaran, V., Taylor, A., and Wang, D. Intersectionality in AI safety: Using multilevel models to understand diverse perceptions of safety in conversational AI. In *Proceedings of the 3rd Workshop on Perspectivist Approaches to NLP (NLPerspectives) @ LREC-COLING 2024*, pp. 131–141, Torino, Italia, 2024. ELRA and ICCL. URL https://aclanthology.org/2024.nlperspectives-1.15/.

Hovy, D., Berg-Kirkpatrick, T., Vaswani, A., and Hovy, E. Learning whom to trust with MACE. In *Proceedings of NAACL-HLT*, pp. 1120–1130, 2013.

Jang, J., Kim, S., Lin, B. Y., Wang, Y., Hessel, J., Zettlemoyer, L., Hajishirzi, H., Choi, Y., and Ammanabrolu, P. Personalized soups: Personalized large language model alignment via post-hoc parameter merging, 2023.

Kim, G. and Kim, E. Swap-guided preference learning for personalized reinforcement learning from human feedback. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://arxiv.org/abs/2603.12595.

Kirk, H. R., Whitefield, A., Röttger, P., Bean, A. M., Margatina, K., Ciro, J., Mosquera, R., Bartolo, M., Williams, A., He, H., Vidgen, B., and Hale, S. A. The PRISM alignment dataset: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Datasets and Benchmarks Track*, 2024. doi: 10.52202/079017-3342.

Kobalczyk, K., Fanconi, C., Sun, H., and van der Schaar, M. Few-shot steerable alignment: Adapting rewards and LLM policies with neural processes, 2024.

Koren, Y. and Sill, J. OrdRec: An ordinal model for predicting personalized item rating distributions. In *Proceedings of the Fifth ACM Conference on Recommender Systems (RecSys)*, pp. 117–124, 2011.

Koren, Y., Bell, R., and Volinsky, C. Matrix factorization techniques for recommender systems. *Computer*, 42(8): 30–37, 2009.

Kumar, D., Kelley, P. G., Consolvo, S., Mason, J., Bursztein, E., Durumeric, Z., Thomas, K., and Bailey, M. Designing toxic content classification for a diversity of perspectives. In *Seventeenth Symposium on Usable Privacy and Security (SOUPS 2021)*, pp. 299–318. USENIX Association, 2021. ISBN 978-1-939133-25-0. URL https://www.usenix.org/conference/soups2021/presentation/kumar.

Li, X., Zhou, R., Lipton, Z. C., and Leqi, L. Personalized language modeling from personalized human feedback, 2024.

Linacre, J. M. *Many-Facet Rasch Measurement*. MESA Press, Chicago, 1989.

Luo, F., Yang, R., Sun, H., Deng, C., Yao, J., Shen, J., Zhang, H., and Chen, H. Rethinking diverse human preference learning through principal component analysis. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 19857–19870, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.1019.

Miranda, L. J. V., Wang, Y., Elazar, Y., Kumar, S., Pyatkin, V., Brahman, F., Smith, N. A., Hajishirzi, H., and Dasigi, P. Hybrid preferences: Learning to route instances for human vs. AI feedback. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7162–7200, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.355.

Mishra, P., Rastogi, C., Pfohl, S. R., Parrish, A., Teh, T. H., Patel, R., Diaz, M., Wang, D., Paganini, M., Prabhakaran, V., Aroyo, L., and Rieser, V. Decoding safety feedback from diverse raters: A data-driven lens on responsiveness to severity. *Transactions on Machine Learning Research*, 2026. URL https://arxiv.org/abs/2503.05609.

Mostafazadeh Davani, A., Díaz, M., and Prabhakaran, V. Dealing with disagreements: Looking beyond the majority vote in subjective annotations. *Transactions of the Association for Computational Linguistics*, 10:92–110, 2022. doi: 10.1162/tacl_a_00449.

Orlikowski, M., Röttger, P., Cimiano, P., and Hovy, D. The ecological fallacy in annotation: Modelling human label variation goes beyond sociodemographics. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 1017–1029, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-short.88.

Park, C., Liu, M., Kong, D., Zhang, K., and Ozdaglar, A. RLHF from heterogeneous feedback via personalization and preference aggregation, 2024.

Plank, B. The "problem" of human label variation: On ground truth in data, modeling and evaluation. In *Proceedings of EMNLP*, pp. 10671–10682, 2022.

Poddar, S., Wan, Y., Ivison, H., Gupta, A., and Jaques, N. Personalizing reinforcement learning from human feedback with variational preference learning. In *Advances*

in Neural Information Processing Systems 37 (NeurIPS 2024), 2024. doi: 10.52202/079017-1664.

Rao, P. V. and Kupper, L. L. Ties in paired-comparison experiments: A generalization of the Bradley-Terry model. *Journal of the American Statistical Association*, 62(317): 194–204, 1967. doi: 10.1080/01621459.1967.10482901.

Raykar, V. C., Yu, S., Zhao, L. H., Valadez, G. H., Florin, C., Bogoni, L., and Moy, L. Learning from crowds. *Journal of Machine Learning Research*, 11:1297–1322, 2010.

Ryan, M. J., Shaikh, O., Bhagirath, A., Frees, D., Held, W., and Yang, D. SynthesizeMe! inducing persona-guided prompts for personalized reward models in LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. URL https://arxiv.org/abs/2506.05598.

Shenfeld, I., Faltings, F., Agrawal, P., and Pacchiano, A. Language model personalization via reward factorization, 2025.

Siththaranjan, A., Laidlaw, C., and Hadfield-Menell, D. Distributional preference learning: Understanding and accounting for hidden context in RLHF. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2312.08358.

Sorensen, T., Moore, J., Fisher, J., Gordon, M. L., Mireshghallah, N., Rytting, C. M., Ye, A., Jiang, L., Lu, X., Dziri, N., Althoff, T., and Choi, Y. Position: A roadmap to pluralistic alignment. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 46280–46302, 2024.

Sorensen, T., Mishra, P., Patel, R., Tessler, M. H., Bakker, M. A., Evans, G., Gabriel, I., Goodman, N., and Rieser, V. Value profiles for encoding human variation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 2047–2095, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-main.106.

Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., Bikel, D., Blecher, L., Canton Ferrer, C., Chen, M., Cucurull, G., Esiobu, D., Fernandes, J., Fu, J., Fu, W., Fuller, B., Gao, C., Goswami, V., Goyal, N., Hartshorn, A., Hosseini, S., Hou, R., Inan, H., Kardas, M., Kerkez, V., Khabsa, M., Kloumann, I., Korenev, A., Koura, P. S., Lachaux, M.-A., Lavril, T., Lee, J., Liskovich, D., Lu, Y., Mao, Y., Martinet, X., Mihaylov, T., Mishra, P., Molybog, I., Nie, Y., Poulton, A., Reizenstein, J., Rungta, R., Saladi, K., Schelten, A., Silva, R., Smith, E. M., Subramanian, R., Tan, X. E., Tang, B., Taylor, R., Williams, A., Kuan, J. X., Xu, P., Yan, Z., Zarov, I., Zhang, Y., Fan, A., Kambadur, M., Narang, S., Rodriguez, A., Stojnic, R., Edunov, S., and Scialom, T. Llama 2: Open foundation and fine-tuned chat models, 2023.

Uma, A. N., Fornaciari, T., Hovy, D., Paun, S., Plank, B., and Poesio, M. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470, 2021.

Wang, Z., Bukharin, A., Delalleau, O., Egert, D., Shen, G., Zeng, J., Kuchaiev, O., and Dong, Y. HelpSteer2-Preference: Complementing ratings with preferences. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2410.01257.

Zhang, L. H., Milli, S., Jusko, K., Smith, J., Amos, B., Bouaziz, W., Revel, M., Kussman, J., Sheynin, Y., Titus, L., Radharapu, B., Yu, J., Sarma, V., Rose, K., and Nickel, M. Cultivating pluralism in algorithmic monoculture: The community alignment dataset. In *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026. URL https://arxiv.org/abs/2507.09650.

Zhang, M. J. Q., Wang, Z., Hwang, J. D., Dong, Y., Delalleau, O., Choi, Y., Choi, E., Ren, X., and Pyatkin, V. Diverging preferences: When do annotators disagree and do models know? In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 76193–76212, 2025.

Zhao, S., Dang, J., and Grover, A. Group preference optimization: Few-shot alignment of large language models. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2310.11523.

Zollo, T. P., Siah, A. W. T., Ye, N., Li, A., and Namkoong, H. PersonalLLM: Tailoring LLMs to individual preferences. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2409.20296.

A. Protocols and amendments

Chronology and evidential status. - **Graded-label stages (1–2c, MultiPref, DICES, PRISM graded, code readability).** Rules were frozen before outcomes. The symmetric offsets DFs were added after the first results and are exploratory. - **Published methods and positive controls (Stages 3, 5, 6).** Rules were frozen. The dose-response analyses are exploratory. - **CA turn 1.** - Stage 8 rule: frozen. - Stages 9–11: exploratory development of the adaptation procedure, after inspecting test results. - Stage 13 S13-1: frozen tests on these previously examined data. Am. 2 and Am. 4 were added after the result. - **CA turn 2 (Stage 12) and CA turn 3 (S13-3).** Frozen procedures, evaluated on outcomes not previously analysed. The users are the same people. - **MultiPref under the CA procedure (S13-2).** Frozen. The MultiPref data had been analysed earlier with other methods.

All intervals are nominal and do not account for the earlier adaptive development.

All analyses were specified in timestamped internal protocols (PROTOCOL_STAGE1, PROTOCOL_STAGE2, PROTOCOL_STAGE2B, PROTOCOL_STAGE2C, PROTOCOL_STAGE3); stages 2b and 2c were written after the stage-2a results. Changes after a result were logged as numbered amendments with the reason; the main ones: the first decoding split (a = b = 2) was structurally empty (two ratings determine their distribution) and was replaced by a = 3, b = 1; hand-set penalties in the rater–item model over-fitted and were replaced by inner cross-validation (which reversed an earlier, wrong conclusion that rater shifts do not help); penalty grids were extended when selections fell on grid edges, before any hypothesis was read; the low-rank baseline lacked a constant column and was fixed in its favour; the probit group model had a scale non-identifiability and was fixed; VPL suffered posterior collapse and its KL weight was re-selected including 0; PAL-B's adaptation ridge was re-selected by k-shot validation. Every superseded full run is kept and reported in the supplementary results; two debugging smoke runs and one fold whose output was deleted unread when the penalty grids were extended are not. The real-data aspect contrast of §5.7 was revised four times: own − other; with a style-only control after a first, confounded +0.45 [+0.01, +0.94]; against $\theta = 0$; and against the population weighting, after the reference learner turned out to shrink towards 0 rather than towards the population. The few-shot-tuned low-rank prior was selected at the top of its grid (8192) in most folds; extending the grid to 524,288 left every DL result unchanged in substance (real data: own − other +0.31 vs +0.30 at k = 20, DFs − DL +0.003 vs +0.0045; synthetic +5/+10-point panels: +0.10 / +0.09 vs −0.02 / +0.09).

B. Hypotheses and outcomes

Stage 2a (MultiPref, k = 20): H1 DF > D — holds (+0.080); H5 (probit F1 > D, the primary probit contrast after Amendment 1) — not supported (+0.020 [−0.001, +0.041]; homoscedastic F − D −0.029), superseded by H1 in Amendment 2; H2 DF > group — holds; H3 DF ≥ low-rank − 0.005 — holds under the training-rater prior, not robust under a few-shot-tuned prior (−0.015); H4 rater scale > rater shift — holds on MultiPref only; H6 symmetric style model gains likelihood without losing direction accuracy — holds and robust. Replications: H1r, H2r hold on DICES and PRISM; H3r holds on DICES (robust) and is a tie on PRISM under the tuned prior. Stage 2b: B1, B2, B3 hold. Stage 2c: C1 holds (graded), C2 holds (binary: |DFs − D| accuracy −0.0002; all personalisation gains ≤ 0.002 nats), C3 fails (published latents do not encode style, $R^2 \approx 0$), C4 not met as a conjunction (first part holds, +0.088 over VPL; second part a tie with DFs, lower bound −0.014); Stage 3 (positive control): $\tau 0$ control passes; power at the prespecified strengths (+1.7, +5.2 oracle points) not established — no content-interacting method's own − other gain has a lower bound above +1 point; exploratory dose-response: VPL and PAL-B pass at +21 points, none at +10; the split-half content estimator (S3-C) has no power; with full features (exploratory, Amendment 5) the $\tau 0$ control passes and PAL-B passes at +10 points (k = 20), so the PCA bottleneck explains part of the detection limit. R1 (PRISM native pairwise choices: every method's own choice-accuracy gain at k = 20 has an upper bound < +0.01) holds (largest bound +0.0071, PAL-B K = 4).

Scale use and the shared reward model. Training the text model jointly with rater offsets and then using the population-mean offsets (k = 0) does not give a better consensus model than ordinary training: −0.003 [−0.010, +0.003] nats on MultiPref (direction +0.3 points, n.s.), −0.022 [−0.037, −0.008] on DICES, +0.001 on PRISM, −0.009 [−0.016, −0.004] with the LoRA reward model. Ordinary training averages scale use out; it matters for predicting a person, not for learning the content preference.

What the offsets encode. In the probit family, rater *scale* matters more than rater *shift* on MultiPref (+0.036 [+0.020, +0.057] at k = 20), but rater *shift* dominates on DICES (−0.053) and PRISM (−0.031): leniency/strictness on single-response rating scales, intensity on two-sided preference scales.

What this does and does not say about pluralism. Our results do not show that people agree about content: the item-level disagreement that replicates across rater panels is real (split-half reliability 0.25 on MultiPref) and about a tenth of it is decodable from text. They show that, for predicting an individual new rater from few ratings, that content-level disagreement is a small part of what is predictable, and that

demographic or crowd/expert groups add nothing beyond the text.

Fine-tuned and in-context reward models (§5.3, full text). With a LoRA-fine-tuned Qwen2.5-7B reward model (D-LoRA, trained per fold on the training raters' rating distributions) as the text model, the three-number style model still adds +0.100 [+0.071, +0.130] nats per rating at k = 20 — the same as its gain on the frozen-feature head (+0.100). Fine-tuning improves the text-only model by +0.027 [+0.016, +0.039] (direction accuracy +1.4 points) and the style model by the same +0.027: the two gains add. Personalising the frozen-feature model with 20 ratings per rater gains more than fine-tuning the text model (+0.072 [+0.041, +0.106]). DFs keeps the direction accuracy of D-LoRA (+0.002 [−0.002, +0.006]). Against in-context personalisation — a separate LoRA model on the same base, trained with ten of the rater's other ratings in the prompt and evaluated with the new rater's k ratings in the prompt — the three offsets on D-LoRA are better at k = 20. The in-context model's result varies with its training seed (two of fifteen runs failed, gaining −0.078 and −0.055 over D-LoRA against +0.021 to +0.125 for the others; the two failed runs have the highest training loss), so it was trained with three seeds per fold, and we fixed in advance that "better" requires a lower bound above zero for every seed and for the seed average. Averaged over seeds — the expected result of one training run — the offsets are better by +0.041 [+0.027, +0.054] nats per rating (209 raters; per seed +0.062, +0.018 and +0.042, every lower bound at least +0.005) and more accurate on which response the rater prefers (+1.9 points [+0.9, +2.7]). The in-context model does learn from the history (+0.051 [+0.024, +0.079] over D-LoRA on average), at several thousand prompt tokens per prediction, but its direction accuracy is below D-LoRA's (−1.7 points [−2.6, −0.8] on average; −0.6 to −3.0 per seed).

Value conflicts (§5.7, full text). The positive control asks what the methods could detect; MultiPref also lets us ask how much there is to detect. Every MultiPref annotation carries the annotator's own helpfulness, truthfulness and harmlessness preferences next to the overall one, so a rater's way of trading these values off is observable. We fit an ordinal model of the overall preference from the annotator's own aspect labels (their values and absolute values) with rater-specific cut-points (scale use) and either a pooled trade-off or a rater-specific trade-off, adapted on k of the rater's training-item annotations and evaluated on held-out items **with the rater's own aspect labels as inputs** — an oracle for the content side. The pooled trade-off weights harmlessness most (6.1, five-fold mean), then helpfulness (4.0) and truthfulness (2.0). With the rater's own aspect labels as inputs the pooled model already gets 97.5 % of directions right (74 % on conflict items): aspect and overall labels are produced together by the same annotator, so this ceiling concerns a small residual and is a property of MultiPref's concurrent labelling.

Even with the rater's own aspect judgements and all of their training ratings, a rater-specific trade-off predicts which response they prefer only 0.1 points better than the population's trade-off, and even fitted in-sample only 0.4 points better. The reason is that the aspects point in opposite directions in only 4.0 % of the held-out non-tie comparisons (3.5 % of all annotations); there individual trade-offs matter a great deal (in-sample +9.7 points; among the 20 raters with at least ten helpfulness–harmlessness conflicts, the share of conflicts resolved in favour of helpfulness ranges from 0.00 to 0.94), but they are too rare to move overall accuracy. Individual trade-offs do change how strongly raters prefer (up to +0.046 nats with all their ratings), in line with the calibration-not-ranking pattern. We fixed in advance that an aspect-weight personalised reward model would be built only if the lower bound of this ceiling exceeded half a point with all training ratings; it does not, so we did not build one. The learned personalisers cannot be tested on the conflict items either: the 243 held-out non-tie ratings on them give own − other intervals wider than the achievable ceiling (k = 20: low-rank latent +0.1 [−2.2, +2.4]). **A calibrated test on the real data.** The ceiling above gives each rater's own aspect labels; a deployable personaliser would know only the item's aspects. We therefore ran the reference learner of §5.6 on the real labels with the item's consensus aspect differences from its *other* annotators as content features (3-d), with a hierarchical prior centred at the population weighting (estimated on the training raters) and compared each rater's own weights with that population weighting (style offsets kept). Individual weights transfer *worse* than the population's: −0.28 [−0.56, −0.06] points at k = 20 and −0.63 [−1.01, −0.25] at k = 50. With injected aspect-weight heterogeneity it gives +0.46 [−0.00, +0.96] at k = 50 when an oracle would gain +1.4 points and +3.4 [+2.6, +4.3] at +5.0. The real data lie well below the +5-point panel and, at k = 50, just below the +1.4-point panel: for this injection family, the real data lie below panels of one to two oracle points (sensitivity evidence, not a validated upper bound). This test is exploratory and post hoc, and its contrast was revised four times (Appendix A; no version shows a positive transferable aspect part), the calibration strengths were chosen after seeing the real bound, it uses one synthetic draw on panels noisier than the real data, and it uses other annotators' aspect labels of the test items (an aspect oracle, not a deployable model). Three caveats: the ceiling bounds personalisation of the *weighting* only — disagreement about which response is more helpful or harmless is absorbed into the inputs; it covers content preferences expressible through these three aspects only; MultiPref's prompt sampling determines how often values conflict; and annotators may have aligned their aspect labels with their

rater ratings used	rater − pooled trade-off, direction accuracy (points)	log-likelihood per rating	on conflict items (points)
k = 20	+0.03 [−0.05, +0.12]	+0.011 [+0.003, +0.020]	+0.6 [−1.1, +2.4]
k = 50	+0.06 [−0.09, +0.21]	+0.023 [+0.011, +0.037]	+0.3 [−2.7, +3.8]
all training items	+0.10 [−0.24, +0.43]	+0.046 [+0.028, +0.068]	+2.0 [−5.0, +9.2]
in-sample (fitted on the test items)	+0.44 [+0.27, +0.64]	+0.094 [+0.079, +0.112]	+9.7 [+5.6, +14.5]

overall label, which would understate conflicts.

Code readability (Stage 10). Pre-set rules: SE-1 (offsets transfer) holds on both panels; SE-2 with the low-rank latent: Buse–Weimer inconclusive (own gain in concordance +1.75 [+0.68, +2.78], own − other +0.53 [−0.63, +1.72]; the text-only model orders snippets at chance, 51.6 %), Dorn degenerate (validation selects the largest penalties, the latent equals the text-only model). SAP-code, own − population / own − other: Buse–Weimer k = 20 +0.40 [−0.25, +1.06] / +1.57 [+0.89, +2.25], k = 50 +1.30 [+0.46, +2.18] / +2.98 [+2.00, +3.91]; Dorn k = 10 −0.19 [−0.81, +0.42] / +2.06 [+1.49, +2.62] — the rule as worded is met on Dorn through own − other, which Stage 5 Am. 4 showed is not on the oracle scale, so content personalisation is not claimed. One within-snippet permutation control gives own − population −0.38 (Buse–Weimer) and −2.09 (Dorn).

Untuned rank-2 comparator (Stage 9 Am. 4). With only its map ridge chosen on training raters' held-out ratings and z ˜ N(0, I) for new raters, the rank-2 attribute model gave −2.67 [−3.30, −2.07] (null), +2.20 / +7.36 (semantic), −0.42 (out-of-dictionary) and −2.13 [−2.73, −1.52] (real); tuning its adaptation strength on pseudo-new raters (Am. 8) removes the losses.

SAP with the pre-registered EM update (Stage 9, before Amendment 7). own − μ at k = 20: null draws −0.28 … +0.05 (k = 50: down to −0.50 [−0.92, −0.09]); semantic panels +2.26 … +2.79 and +7.30 … +8.04; out-of-dictionary +0.41 and +0.81; real MultiPref +0.17 [−0.19, +0.55], LL +0.0022 [+0.0003, +0.0041] nats; PRISM native choices −0.25 [−1.30, +0.83] (EM variances stayed near their initial value there; with the fixed-point update +0.08 [−0.86, +1.02]).

Positive control on full features (Stage 3, Amendment 5). Because the compressed features might be what limits the content-interacting methods, we re-ran VPL and PAL-B on the full features (exploratory). The text-only fits improve (k = 0 log-likelihood: PAL-B −1.533 vs −1.558, VPL −1.495 vs −1.574), but VPL's latent then collapses and is unused at every strength (own − other exactly 0), so this check rests on PAL-B. Its detection limit drops: it passes the threshold at +10 points; at +5 points its k = 50 gain is positive (+1.05 [+0.37, +1.72]; real data +0.19 [−0.12, +0.48]) but meets the +1-point criterion in only one of four synthetic draws. The real data stay at the null level (+0.11 [−0.16, +0.37] at k = 20). With this stronger detector, the real data lie below the +10-point panels of the injected form at k = 20; at k = 50 they lie below the +5-point panel but the intervals overlap (comparisons with synthetic panels, not formal exclusion bounds); the compressed features used for the main tables make the published methods weaker detectors than they need to be.

Positive control (§5.6, full text). No method has power at the pre-specified strengths. In the exploratory dose-response, VPL and PAL-B detect content preferences worth +21 points but recover less than half a point of +10; the low-rank latent, whose user term interacts with a joint embedding of the comparison, has a signal at +21 points that stays below the threshold, and with the untuned prior it shows nothing up to +5 points either. On the real data VPL and PAL-B are at the null-control level, and the low-rank latent's marginal +0.30 [−0.00, +0.60] is matched by the text-independent offsets (+0.32 [+0.06, +0.58] on the same contrast), so it reflects scale use (which can tip a few near-tie predictions), not content. Read together: on MultiPref, the real data lie far below the +20-point panels for these methods, while preferences worth up to about 10 points cannot be seen by them from 20 ratings, so the ranking null of §6 says nothing about those. These are comparisons with synthetic panels of one injected form, not formal exclusion bounds. On the full features (exploratory) VPL's latent collapses (own − other exactly 0), while PAL-B passes the threshold at +10 points but not at +5, and the real data stay at the null level (+0.11 [−0.16, +0.37]): the real data lie below the +10-point panels of the injected form at k = 20 but not below the +5-point panels (Appendix B). **How far are these methods from what the data allow?** A well-specified reference learner — the injection's two features known, a per-rater weight vector with an empirical-Bayes prior, style offsets re-fitted, the same permutation contrast — recovers the injected preference from 20 ratings at every strength. On the oracle's scale (the rater's own weights against the population state, $\theta = 0$ here, with the rater's style offsets kept) it gains +0.89 [+0.40, +1.37] points at +1.7 oracle points, +3.1 [+2.3, +4.0] at +5.2 and +8.0 at +10.1, i.e. 52–79 % of the oracle; own − other roughly doubles these numbers, because another rater's weights are anti-informative, and is therefore not used for this learner. Among the tested strengths, the reference learner crosses the detection criterion between +1.7 and +5.2 points, well below the methods'. Is the gap in the methods' inputs or in the learners? Linear probes (exploratory, Appendix B) show the injected direction is available to VPL and PAL-B (out-of-fold R^2 0.65 compressed, 1.00 full) but not to the

low-rank latent's joint embedding (0.02); re-injected inside that embedding, the latent detects +10 points (+1.97 [+1.31, +2.65]; originally +0.09; 3 of 4 draws) but not +5 (+0.08; 0 of 4). With the dimension available, the best learners thus detect between +5 and +10 oracle points, the reference learner below +5: what fails is finding the dimension, not the number of ratings. With semantic axes (formatting, hedging/refusal) no method detects even +9.7 points; the reference learner recovers +7.4 (Appendix B). Results are stable over four synthetic draws, and the synthetic panels are noisier than the real data, so the calibrations transfer only approximately (Appendix B).

Positive control: seed variability and realism (Stage 5). Over four synthetic draws (θ and labels) the reference learner is stable (+2.4 to +3.1 at +5 points, +6.4 to +8.0 at +10), PAL-B on full features detects the +10-point panels in all four (lower bounds +1.16 to +3.23) and the +5-point panels in none (+0.26 to +1.40), and the low-rank latent detects neither in any draw (-0.31 to +0.15). A limitation of all synthetic panels: they are noisier than the real data (rater-pair direction agreement 0.59 at $\tau = 0$ against 0.673 [0.666, 0.680]; Krippendorff's α 0.11 against 0.28), and adding item fixed effects to the generator did not close the gap (0.60), nor does it reproduce the small effect of style offsets on direction (+0.32 real vs -0.07 at $\tau = 0$); the calibrations transfer to the real data only approximately, with no claimed direction of bias.

LoRe on all panels (Stage 6). Across PRISM pairs and MultiPref, binary and graded labels and k $\in \{5, 20, 50\}$, LoRe never improves direction or choice accuracy over its own k = 0 version or over the text-only model (own gain -6.2 to +0.1 points; the regularised version -1.1 to -0.0), and its log-likelihood is below the text-only model's everywhere. Its own $-$ other is positive in some cells (MultiPref binary, k = 20: +1.3 [+0.5, +2.0]) while its own gain is negative there (-1.2 [-1.9, -0.5]): another rater's over-fitted weights are anti-informative, the scale problem discussed for the reference learner, so own $-$ other alone is not evidence of personalisation for methods whose few-shot fit over-fits.

Stage 7 (after the seventh review; PROTOCOL_STAGE7.md). S7-1, cross-fitted ridge probes (prompt-group folds, out-of-fold R^2) of the two injected features from each learner's inputs: log length ratio 0.88 (VPL/PAL-B PCA-256 x_B $-$ x_A), 0.93 (full features), 0.73 (D/DL joint PCA-256); random direction 0.65, 1.00, 0.02. By the rule fixed beforehand, S7-2 was run for the low-rank latent only: the random direction re-drawn inside its joint embedding (orthogonal to the consensus direction), τ calibrated to oracle gains of +5.1 and +10.4. Own $-$ other at k = 20 / 50: +0.08 [-0.18, +0.33] / +0.03 [-0.29, +0.34] at +5.1; +1.97 [+1.31, +2.65] / +7.02 [+5.84, +8.16] at +10.4; the reference learner is unchanged (own $-$ population +2.67 and +8.33). S7-3, seen-rater ceiling on real MultiPref (training raters $\times$ test items, 904 rater-folds, trained parameters; derangement of training raters): direction accuracy minus the text-only model -0.51 [-0.88, -0.11] (five offsets), -0.00 [-0.20, +0.18] (symmetric), -0.42 [-0.81, -0.02] (low-rank); own $-$ other -0.23, +0.10, -0.11 [-0.57, +0.36]; log-likelihood over the text-only model +0.13 nats for all three. By the rule fixed beforehand, the ranking null also holds for seen raters. S7-4 (Amendment 3), semantic axes — (s1) log(1 + markdown list/heading lines) of B minus A, (s2) the same for a fixed list of hedging/refusal phrases — calibrated to +4.6 and +9.7 oracle points: probes R^2 0.37–0.66 (formatting) and 0.09–0.24 (hedging); own $-$ other at k = 20 / 50 for the low-rank latent +0.09 / +0.05 and -0.18 / -0.08, VPL -0.05 / +0.04 and +0.03 / +0.03, PAL-B (compressed features) -0.24 / -0.12 and +0.27 / +0.40 [-0.02, +0.85]; reference learner own $-$ population +2.57 [+1.61, +3.57] and +7.44 [+5.96, +8.91]. No method detects either panel. S7-5 (Amendment 4), the in-span panels over four draws (θ and label seeds 1, 11, 12, 13): the low-rank latent's own $-$ other at k = 20 is +1.97, +2.83, +3.14, +1.44 [+0.92, +1.96] at +9.3 to +10.9 oracle points (detected in 3 of 4; k = 50 +5.0 to +7.3) and +0.00 to +0.14 at +4.7 to +5.5 (0 of 4); the reference learner +7.4 to +8.3 and +2.6 to +2.9. Both statements are kept under the rule fixed beforehand.

Confirmatory and exploratory results. Confirmatory (rule fixed before the result): the stage-2a hypotheses H1–H5 and their replications (H6, on the symmetric offsets, was added after the first results and is exploratory; its later replications were evaluated prospectively); B1–B3; C1–C4; R1 (PRISM native choices); the Stage-3 positive control at +1.7 and +5.2 oracle points (no power); the Stage-4 ceiling rule (not met); the Stage-6 LoRe rule (PRISM null not withdrawn). Exploratory (added or changed after a result, labelled in the text): the symmetric offsets DFs and their hypothesis; the few-shot-tuned prior; the +10 and +21 panels and the full-feature runs; the reference learner and its oracle-scale contrast; the calibrated real-data aspect test; the regularised LoRe; the two-way bootstrap re-estimates.

Reproducing LoRe's PRISM result (Stage 6, Amendments 2–3). Unseen-user binary accuracy, mean $\pm$ sd over five random user splits, LoRe's native setting (users with at least six dialogues split 50/50, half of each user's conversations for few-shot and half for test, one chosen-vs-other pair per conversation; 1,267 users, about three few-shot pairs per unseen user):

Taking one pair per interaction instead (every turn; 20 pairs per user) gives 58.8 / 59.2 / 57.0 with our features. The remaining differences — a stronger monolithic baseline here

embeddings	monolithic BT	LoRe, no adaptation	LoRe, few-shot fit
Skywork-Reward-Llama-3.1-8B-v0.2 (LoRe's)	67.4 ± 1.1	67.3 ± 1.8	63.6 ± 3.7
Qwen2.5-7B-Instruct (ours)	67.7 ± 1.5	68.7 ± 1.7	61.0 ± 2.5
published (Bose et al., 2025)	61.0 ± 1.2	—	71.0 ± 1.0

and a few-shot fit that helps there but hurts here — could not be traced to any setting the LoRe paper reports. Under our protocol (user- and conversation-disjoint folds) the Skywork embeddings leave the conclusion unchanged: LoRe's own gain at k = 20 is −1.3 [−2.5, −0.2] points (regularised −0.5 [−1.3, +0.2]) and own − other +0.2 [−0.9, +1.2], as with our features.

Positive-control details (moved from §5.6). In the exploratory dose-response, VPL and PAL-B detect +21-point preferences but recover less than half a point of +10; on the real data they are at the null-control level, and the low-rank latent's +0.30 [−0.00, +0.60] is matched by the text-independent offsets (+0.32), i.e. scale use. With full features PAL-B detects +10 points but not +5. **How far are these methods from what the data allow?** A well-specified reference learner — the two injected features known, an empirical-Bayes prior on each rater's weights, style offsets re-fitted — recovers +0.89 [+0.40, +1.37], +3.1 [+2.3, +4.0] and +8.0 points of +1.7, +5.2 and +10.1-point oracle gains from 20 ratings (own versus population weights, the oracle's scale). Linear probes show the injected direction is available to VPL and PAL-B (out-of-fold R^2 0.65 compressed, 1.00 full) but not to the low-rank latent's joint embedding (0.02); re-injected inside that embedding, the latent detects +10 points (3 of 4 draws) but not +5 (0 of 4). With semantic axes (formatting, hedging/refusal) no method detects even +9.7 points. Twenty ratings permit useful recovery in these injected families when the generating features are supplied; the observed gap implicates feature representation and the learners' inductive biases. These are comparisons with synthetic panels of specific injected forms (noisier than the real data), not formal exclusion bounds; details and seed variability in Appendix B. Generator and calibration as first described: We keep MultiPref's structure (raters, items, who rated what, folds and k-shot draws) and replace the labels by semi-synthetic ones. A generator — the text-only head plus scale-use offsets, fitted on all ratings — gives each rating a distribution, to which a rater-specific content term $(\theta_r^\top g_i)\,(-2,-1,0,1,2)$ is added to the logits; g_i are two features of the response pair (the log length ratio and a random direction in the per-response embedding space, orthogonal to the consensus quality direction) and $\theta_r \sim N(0, \tau^2 I)$. The scale τ is calibrated on the generator alone so that an oracle that knows θ_r gains +1.7, +5.2, +10.1 or +21.1 points of direction accuracy on the test ratings; $\tau = 0$ is a null control. Every method is then run under the paper's protocol. Because a method whose unpersonalised state is not the population state can improve with anyone's ratings, the contrast is **the method adapted on the rater's own k ratings minus the same method given another test rater's adapted parameters** (a random derangement within fold, k and draw). Before the runs we fixed that the protocol has power at a strength if some content-interacting method has a lower 95 % bound above +1 point at k = 20; the +10 and +21 panels and the untuned-prior low-rank runs were added after the first results and are exploratory.

5.7 How much room is there? (full text moved from §5). Every MultiPref annotation carries the annotator's own helpfulness, truthfulness and harmlessness preferences, so a rater's trade-off between these values is observable. With the rater's own aspect labels as inputs (an oracle for the content side), a pooled trade-off already gets 97.5 % of directions right, and a rater-specific trade-off fitted on all of the rater's training ratings adds only +0.10 [−0.24, +0.43] points (+0.44 even in-sample). The three recorded aspect judgements point in opposite directions in only 4.0 % of held-out non-tie comparisons; there individual trade-offs matter (in-sample +9.7 points), but they are too rare to move accuracy. A calibrated test with the item's consensus aspects from other annotators finds individual weights transferring worse than the population's (−0.28 [−0.56, −0.06] at k = 20; exploratory, Appendix B).

5.9 Developers rating code readability (full text moved from §5). Two code-readability studies have persistent developer identities: Buse–Weimer (121 raters × 100 Java snippets) and Dorn (3,393 developers with at least ten ratings of 360 Java/Python/CUDA snippets). With Qwen2.5-Coder-7B features, scale use transfers to new developers: offsets gain +0.054 [+0.036, +0.074] nats per rating at k = 20 (Buse–Weimer) and +0.040 [+0.031, +0.049] at k = 10 (Dorn). For content we measure within-developer concordance of predicted and given ratings over snippet pairs, comparing a developer's own weights with population weights under the same offsets. SAP with 22 classical readability attributes (line and identifier length, indentation, nesting, comments, operators, ...) gives +0.40 [−0.25, +1.06] points at the pre-set k = 20 on Buse–Weimer (+1.30 [+0.46, +2.18] at k = 50, exploratory) and −0.19 [−0.81, +0.42] on Dorn; the low-rank latent is inconclusive on Buse–Weimer and switched off by validation on Dorn. Details, the pre-set rule's literal outcome on Dorn (met through the own − other contrast, which is not on the oracle scale) and permutation controls are in Appendices A–B.

C. Implementation

Rater identities. All four panels carry persistent rater identifiers (MultiPref pseudonyms, 227 evaluators, median 166 ratings, never twice on one item; DICES rater IDs, 123 raters × 350 items in the release, 104 after its quality filter (used throughout); PRISM user IDs, one survey record each; Community Alignment annotator IDs). Demographics never vary within an ID on DICES and in fewer than 0.3 % of IDs on Community Alignment. On MultiPref, which has no demographics, an evaluator's tie rate and extreme-rating rate in the first and second half of their ratings (by timestamp) correlate at 0.82 and 0.78 across evaluators, against -0.03 and 0.00 (largest 0.17) with evaluator labels shuffled (`check_rater_ids.py`). Shared accounts cannot be ruled out.

Text features: last-token final-layer states of Qwen2.5-7B-Instruct under a 5-point judge prompt (MultiPref, PRISM) or a 3-level safety prompt (DICES), label-free PCA to 256 dimensions. Folds: 5 rater folds (stratified by group) × 5 prompt-group folds; validation: 15 % of training prompt groups. k-shot draws: 5 per k, identical across all models (RC-LoRA: draw 0 only). PRISM item folds are by conversation. Bootstrap: rater clusters, B = 2,000. LoRA: r = 16 on all attention and MLP projections, lr 10^{-4}, one epoch, accumulation 8. VPL / PAL-B: official architectures (latent 512, hidden 512; d_pref 512, mlp2-gelu projectors, angle learner) with an ordered-logistic link for graded labels; VPL's KL weight selected from $\{0.01$ (official), $0.001, 0\}$ and PAL-B's adaptation ridge by k-shot validation (non-official choices made to prevent collapse and over-fitting).

D. Where is the predictable structure? Estimators and results

Estimators. For an item with ratings on K ordered levels, let h be the empirical rating distribution and C the cumulative operator. Split the item's raters at random into disjoint panels P_1 (a raters) and P_2, and let q be a cross-fitted *location-only* forecast with exactly the mode and mean of P_1 (built from other items in the same mode-and-mean cell). With $r_1 = C(h_1 - q)$, $r_2 = C(h_2 - q)$, $M = \|r_1\|^2/n$ and $X = \langle r_1, r_2\rangle/n$, the forecasting value of the raw distribution over its own location obeys the identity $S(h_1) - S(q) = M - 2X$ under the ranked probability score, so $\rho = X/M$ is a split-half reliability of the distribution's *shape* and the raw distribution beats its location exactly when $\rho > 1/2$. Text decodability is the proper-score gain $G = (2\langle \hat{r}, r_2\rangle - \|\hat{r}\|^2)/n$ of a residual $\hat{r}$ decoded from frozen text features, cross-fitted over prompt groups, reported as the fraction G/X of the reproducible shape. The rater–item decomposition fits ordered-probit models $Y^*_{ir} = \theta_i + \delta_r + \sigma_i\sigma_r\varepsilon$ with item locations θ_i, rater shifts δ_r, rater scales σ_r and item dispersions σ_i, compares held-out likelihood over random cells (every held-out rater and item appears in training), and chooses all penalties by inner cross-validation.

Reproducibility. With the split-half estimator defined above, the reliability of the rating distribution's shape beyond its exact mode and mean is ρ = 0.25 [0.22, 0.29] on MultiPref overall preference with three raters (helpful 0.29, harmless 0.21): the raw few-rater distribution forecasts a new rater worse than its own location ($\rho < \frac{1}{2}$). **Text decodability.** A cross-fitted ridge decoder on frozen Qwen2.5-7B-Instruct features, scored on an independent rater panel, recovers G/X = 10.5 % [5.7 %, 11.9 %] of the reproducible shape (within-cell permutation control ≤ 0; positive control 0.35) — comparable to the upper end for ordinal quality ratings (upper bounds 0.4–10 %) and below NLI (17–26 %). **Rater vs item.** A long-format ordered-probit decomposition with real rater identities and penalties tuned by inner cross-validation: rater-specific thresholds improve held-out likelihood by +0.036 nats per rating over item location alone, item-specific dispersion by a further +0.0014 (ratio 25; 19–36 across the five folds); of the rater part, rater scale contributes +0.031 and rater shift +0.005.

E. Protocols and what did not hold

All analyses are documented in an analysis log: timestamped internal protocols written before each stage (not public registrations, so they document rather than certify the order of decisions); changes after results are logged as amendments with reasons, and the symmetric style model DFs and its hypothesis were added after the first full result. Not supported or not robust: (i) the stage-1 decision rule returned "mixed" (text decodability not ≤ 10 %); (ii) non-inferiority of the 5-parameter DF to the low-rank latent (DF − DL ≥ -0.005) holds with the training-rater prior but not with a few-shot-tuned prior on MultiPref (-0.015) and is formally not met on PRISM (bound -0.006); (iii) "scale beats shift" holds only on MultiPref; (iv) the originally registered probit-family hypothesis is not supported (F1 − D = +0.020 [-0.001, +0.041], the primary probit contrast after Amendment 1; homoscedastic F − D = -0.029 [-0.049, -0.008]; superseded when the primary family was moved to the distributional head); (v) the prediction that published user latents encode style failed ($R^2 \approx 0$); (vi) the hybrid's non-inferiority to DFs is formally not met (bound -0.014). Four baseline problems found during the study were fixed in favour of the baselines — a missing constant column in the low-rank latent, a scale non-identifiability in the probit group model, VPL's posterior collapse, and PAL-B's near-unregularised adaptation — and the runs before each fix are reported.

F. Community Alignment analysis log (Stages 8, 11, 12)

The full sequence of analyses behind §6, in order, with the corrections of the fifth review applied (event-level simulation as the principal achievable-gain benchmark, the pair-level one as a consistency check; Holm as an approximate sensitivity analysis). Protocols: PROTOCOL_STAGE8.md (Am. 1–2), PROTOCOL_STAGE11.md (Am. 1–10), PROTOCOL_STAGE12.md.

On a dataset built for preference diversity, user-specific signal exists; estimated choice gains remain small. Community Alignment (first turns, 2,957 new users in five countries, the chosen response paired with each other one; Stage 8): at k = 20 every method's own gain over its own k = 0 version has an upper bound below +0.5 points (VPL +0.15 [+0.03, +0.27], LoRe-reg +0.35 [+0.25, +0.46], LoRe -2.34 [-2.79, -1.88]), yet a user's own adapted parameters beat another user's — the frozen rule's own − other half, which a summary-script bug first dropped and which therefore literally reads "content personalisation detected": LoRe +3.27 [+2.85, +3.70], LoRe-reg +0.36, VPL +0.31. A within-country derangement (added after the result) retains the contrast, so country and language do not explain it (own − same-country other +3.22 [+2.83, +3.60]; same-country − any-country +0.03 [-0.12, +0.18]). A cross-fitted, simulation-calibrated diagnostic agrees on what the signal is: users' deviations from the population weights on response length and paragraphing reproduce across disjoint prompt sets (transferable variances 0.32 [0.25, 0.40] and 0.72 [0.23, 1.29]; 0.33 and 0.76 [0.26, 1.35] after prompt-type adjustment; exploratory, Stage 11; with 200 simulation replicates the length false-detection rate is 5.0 % [2.4, 9.0] and paragraphs' power 85 % [79, 89], so neither calibration target is established beyond binomial error), while no method converts them into a one-point gain at k $\leq$ 50. Users' stable prompt mixes could create such residual structure without different preferences on the same tasks; as an observational check the length cross-covariance centred within language × prompt-cluster cells is 0.206 [0.183, 0.231] against 0.210 uncentred, and a same-language user with a matched task mix (cluster shares within L1 0.002) is a worse predictor than the user's own coefficient by +0.80 [+0.45, +1.16] points and +0.009 nats, though better than a random other by +0.75; the two contrasts approximately partition the unmatched own − other (+1.54), which is not a causal decomposition, and the matched other is about 0.35 points worse than the population model (Stage 11 Am. 8). Two annotators agree on 64.3 % of shared response pairs, the text-only model is right on 71.7 % of them. **Using the transferable covariance as the prior** ($\hat{\theta}$_u $-$ μ = c $\hat{C}$ (I + c I_u $\hat{C}$)$^{-1}$ s_u over length, paragraphs and list/heading, $\hat{C}$ estimated on training users only; S11-3) gives +0.13 [-0.28, +0.53] points at c = 1, although the fitted Gaussian model implies +2.8 for this learner at k = 20 (oracle +4.2; with event-level simulated labels that keep one winner per choice event, +1.9 and oracle +3.6 — a third lower, just inside the pre-set factor-1.5 tolerance, Stage 11 Am. 6). The gap is not mysterious: on the evaluated split transfer from support to test prompts is established for length (0.41 [0.27, 0.55]) but not for paragraphs or lists (intervals cover 0), heavier-tailed or concentrated deviations with the same covariance lower the simulated gain of this learner to +1.6 to +0.4 (event-level: +0.9 to -0.4) (so covariance alone does not determine learnability; this does not show that concentration explains the real data), and the full-size prior over-adapts. With the attribute set and prior scale chosen on pseudo-new training users (c = 0.125–0.25 in every fold, never "no adaptation"), transfer-calibrated adaptation gains **+0.49 [+0.20, +0.77] choice points and +0.0088 [+0.0061, +0.0116] nats** on the frozen test users, ahead of the tuned rank-2 model by +0.30 [+0.04, +0.56] points — but not of ordinary validated shrinkage: an isotropic prior over the same three attributes, its scale chosen the same way (τ^2 = 0.1 in every fold), gains +0.66 [+0.37, +0.95] points and +0.0110 nats, and TRV − isotropic is -0.17 [-0.29, -0.05] (not significant after multiplicity adjustment). No advantage of the estimated covariance is established; what helps is the choice of attributes and strong shrinkage (light adaptation). A prompt-group × user two-way bootstrap widens TRV's interval to [+0.01, +0.97]. These are below the one-point threshold, at a different budget (20 choice events, three dependent pairs each) from Stage 8's 20 pairs, where VPL and LoRe-reg also show small positive own gains. **Frozen evaluation on new prompts (Stage 12).** TRV, unchanged, on the same users' second turns (83,238 never-analysed choices; 1,702 test users at k = 20 events): **+0.38 [+0.13, +0.64] choice points and +0.0046 [+0.0031, +0.0063] nats per pair** over μ; validated isotropic shrinkage +0.31 [+0.05, +0.58] and +0.0038; TRV − isotropic +0.07 [-0.07, +0.22]; the tuned rank-2 model selects no adaptation in three of five folds (-0.05). By the pre-set rules the benefit is reproducible (both lower bounds above 0) and sub-threshold (accuracy upper bound below +1), with no advantage over ordinary shrinkage; the prompt-group × user interval for accuracy covers 0 ([-0.04, +0.83]). No language reaches significance alone. The length cross-moment again reproduces across disjoint prompt halves in every fold (0.12–0.19), the list/heading one does not (negative in two folds). These are new prompts and turns of previously studied users, not new users. **Multiplicity (Am. 10; an approximate, exploratory sensitivity analysis: p-values recovered from reported intervals by a symmetric normal approximation, questionable for heavy-tailed cross-moments, then Holm at 0.05).** In Stage 8's frozen family (six methods × two contrasts) the own − other contrasts of LoRe, LoRe-reg, VPL and PAL4 survive, and the detection is unchanged; the positive own gain of VPL (+0.15) and the negative own effects

of PAL4 and DFp do not. Across a family of 29 selected Stage-11 contrasts (not every reported contrast; e.g. AUC contrasts, the k = 32 within-cluster and matched − cyclic contrasts are omitted), the length statements survive (transfer, within-cluster transfer, own − matched other, TRV and isotropic gains, WIMHF-style length), while paragraphs' transferable variance, TRV − rank-2, TRV − isotropic and the new-user length gain of S11-1 (b) (+0.45) do not; the first-turn evidence that isotropic shrinkage beats TRV is therefore weaker, but no advantage of TRV is established either. **Status:** the original Stage 11 diagnostic failed its pre-specified calibration criterion, leaving all three primary outcomes inconclusive and the adaptation gate unmet; all subsequent analyses are exploratory; the final procedure selected attributes and shrinkage on training users only, but its candidate family was developed after inspecting test results, and the bootstrap intervals are nominal and do not account for this adaptation. $\hat{C}$ is an uncentred cross-moment of one-step deviations, not a bounded share of the fitted random-slope variance; the principal achievable-gain benchmark is the event-level simulation (one winner per choice event, Am. 6); the pair-level figures are a consistency check, and both are model-implied gains for specified learners, not general learnability ceilings.

G. Screened attribute personalisation (formerly §5.8)

§5.6 locates the failure: 20 ratings suffice when the content axes are known, and the published learners do not find them. **Screened attribute personalisation (SAP)** supplies candidate axes and learns from the training raters which of them people differ on. Each comparison is described by 24 interpretable attribute differences fixed in advance (length, markdown structure, hedging/refusal, code, digits, questions, pronouns, sentence and word length, examples, disclaimers, summaries, evidence phrases, politeness, refusal, lexical diversity, overlap with the prompt, …); a rater's content term $(\theta_r^\top a_i)(-2,\dots,2)$ is added to the scale-use model's logits, with $\theta_r \sim N(\mu, \operatorname{diag}\sigma^2)$, the population weights μ and the per-attribute variances σ^2 estimated on the training raters by automatic relevance determination (Laplace approximation, MacKay's fixed-point update); a new rater's θ_r is their MAP from k ratings. The contrast is the rater's own weights against the population weights μ, with the rater's own scale-use offsets kept (oracle scale).

Own minus population weights, direction accuracy at k = 20 in points, 95 % rater bootstrap (ranges over four synthetic draws; VPL / PAL-B / low-rank latent: own − other on the same panels, §5.6). Rules fixed in advance: no detection on the null draws (none), detection of the semantic panels in at least three of four draws (four of four each). The SAP column uses the fixed-point variance update adopted after the pre-registered EM update was found to stall when raters have few ratings (EM results in Appendix B); the shared-variance column uses the same update; the rank-2 model's map ridge and adaptation strength (including none) are selected on held-out training raters at k = 20. Exploratory: the ablations and the out-of-dictionary panels.

SAP recovers what the published learners miss and matches the learner told the features; attribute-specific variances improve recovery over one shared variance under the same update (which loses 60–76 % of the gain); a rank-2 subspace on the same attributes, tuned with the same care, matches SAP on the semantic panels and abstains on the null (untuned, it lost 2.1–2.7 points; Appendix B), while on real MultiPref SAP is ahead of it by +0.41 [+0.15, +0.66] points (paired, exploratory). The variance assigned to correlated attributes means the high-variance set is a set of candidates, not independently identified preference dimensions. SAP learns transferable shrinkage over supplied axes; it does not discover new ones (outside the dictionary it recovers under a tenth of a +10.5-point preference). With the fixed-point update its estimated effects on the four null draws are near zero (-0.03 … $+0.11$ at k = 20 and 50; with EM it lost up to 0.5 points at k = 50); this is not a calibrated false-positive rate, and forced adaptation does hurt on Community Alignment (below). On real MultiPref its gain over population attribute weights (the rater's offsets kept) is +0.28 [-0.00, +0.58] direction points and +0.0028 [+0.0013, +0.0043] nats per rating — SAP's achieved gain under this protocol, not a bound on what these attributes could offer. **The decisive test** was frozen before its data were embedded: Community Alignment's native choices (1,897 new users in five languages, k = 20 choice events, per-language population weights, adaptation strength of every method chosen on held-out training users). The fitted model assigns substantial training-user variation to length, paragraphs and formatting (prior variances 0.29–0.37 for length, 0.25–0.30 for paragraphs, 0.19–0.23 for list/heading lines; one fold assigns 1.37 to URLs, ≈ 0 in the others), yet adaptation from 20 choice events does not improve held-out prediction: validation switches SAP off in all five folds, forcing it costs -0.40 [-0.87, +0.06] choice points and -0.0097 [-0.0143, -0.0052] nats, and the tuned low-rank model gains +0.17 [-0.17, +0.51]. The pre-set rule (lower bound above +1 point) is not met.

panel (oracle gain)	SAP	one shared variance	same attributes, rank-2 subs
null, 4 draws (0)	−0.03 … +0.04	+0.06	0 (abstains)
formatting + hedging (+4.6), 4 draws	+2.29 … +2.82	+0.69 [+0.21, +1.18]	+2.78 [+1.84, +3.72]
formatting + hedging (+9.7), 4 draws	+7.44 … +8.12	+3.09 [+2.13, +4.02]	+7.75 [+6.09, +9.36]
two axes outside the dictionary (+10.5)	+0.95 [+0.50, +1.39]	—	—
real MultiPref	+0.28 [−0.00, +0.58]	+0.28 [+0.04, +0.53]	−0.14 [−0.51, +0.24]
real MultiPref, ratings permuted within item	−0.09 [−0.23, +0.05]	—	—
Community Alignment, frozen, k = 20 choice events	0 (abstains); forced −0.40 [−0.87, +0.06]	−0.70 [−1.15, −0.25]	+0.17 [−0.17, +0.51] (tunec